

Nocturnal Visual Place Recognition for Campus-Scale Robot Localization

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Background



Day reference



Night query

- We study **day-to-night visual place recognition** for robot localization in GPS-denied environments.
- The challenge is not only illumination shift, but also **viewpoint change** and live deployment constraints.
- A practical system should build a map once, then localize live camera frames against it reliably.

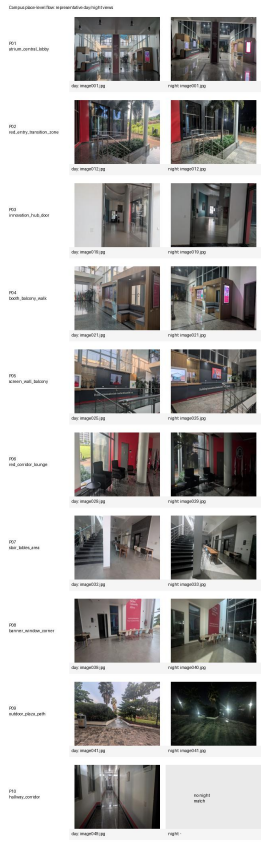
Goal: recognize place from a live camera feed

Methodology

Evaluation Protocol

- Benchmark:** GardensPoint for controlled day-to-night testing.
- Campus set:** 50 day reference images, 64 night queries.
- Metrics: **AUC**, **Recall@1/5/10**, and **Recall@100P**.

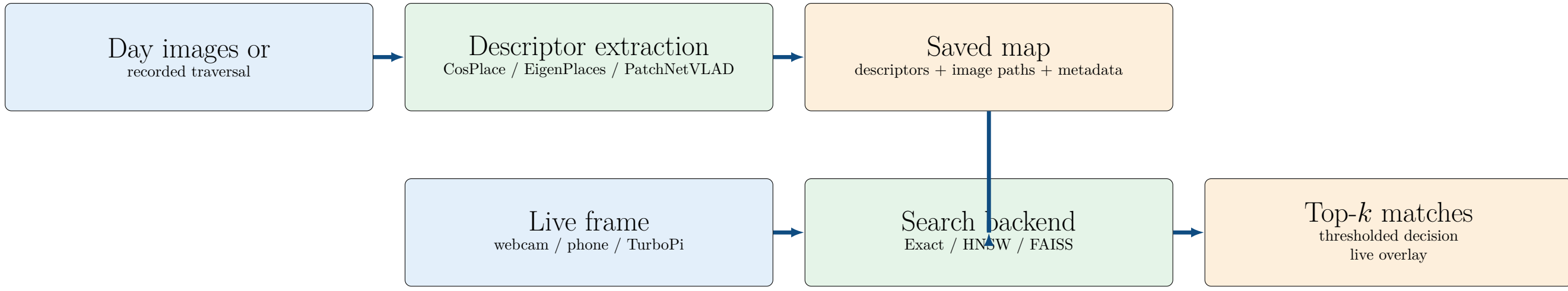
Relabeling Insight



- Original protocol: one night image → one exact day image.
- New protocol: one night image → one **place ID** with multiple valid day views.
- This exposed that strict image-level labels were under-crediting correct same-place matches.

Our Solution

Two-Phase VPR Pipeline



Map Building

- Build from image folders or a recorded traversal video.
- Sample frames, extract descriptors, normalize, save reusable map.

Online Localization

- Process live frames at configurable rate.
- Retrieve top-*k* candidates and apply thresholded recognition.

Applied CV Focus

- Config-driven pipeline for demos.
- Works with webcam, phone webcam, and robot stream.

Offline map building

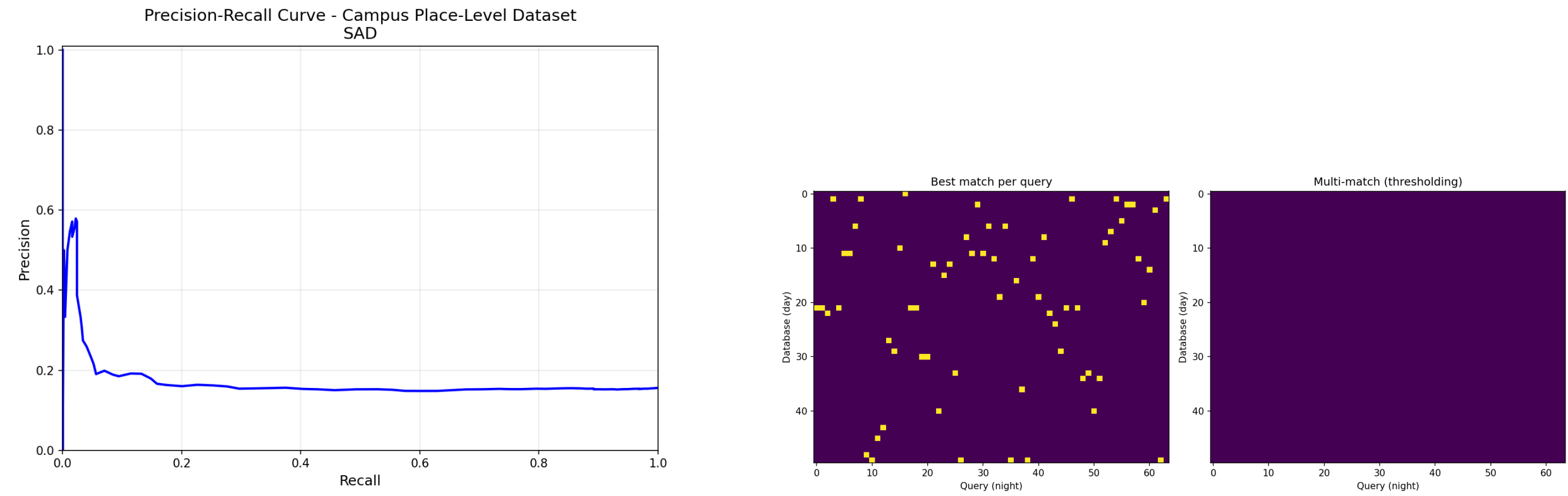
Live retrieval

Robot-facing deployment

Results

Main Result

Original strict labels under-credited multi-view same-place matches



Model	AUC	R@1	Takeaway
CosPlace	0.627	0.969	Best top-1 retrieval after place-level relabeling
EigenPlaces	0.663	0.922	Best overall AUC on campus
PatchNetVLAD	0.641	0.938	Strong place-level alternative
HDC-DELF	0.507	0.922	Much cleaner threshold behavior

- Example gains after relabeling: **CosPlace R@1: 0.727 → 0.969**, **NetVLAD R@1: 0.394 → 0.891**.
- The original setup was not robust to multiple valid views *under a strict image-level evaluation protocol*.

Future Research

Sequence-aware localization: use temporal consistency instead of frame-by-frame decisions only.

Potential filtering: incorporate motion and unselected observations, e.g. Kalman-filter-style prediction.

- Open-set evaluation:** add a dedicated split with truly unmatched night queries after place-level review.
- Robot deployment:** tighten TurboPi integration and evaluate full end-to-end live localization.

References

References

- Berton et al., “Rethinking Visual Geo-localization for Large-Scale Applications,” CVPR 2022. (CosPlace)
- Berton et al., “EigenPlaces: Training Viewpoint Robust Models for Visual Place Recognition,” ICCV 2023.
- Arandjelović et al., “NetVLAD: CNN Architecture for Weakly Supervised Place Recognition,” CVPR 2016.
- Milford and Wyeth, “SeqSLAM: Visual Route-Based Navigation for Sunny Summer Days and Stormy Winter Nights,” ICRA 2012.